

Issuer
Advance

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Inquiry

Project Line 4 OP70

Reference: Line 4 OP70

ID: 012035

Contact persons:

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Address:

Advance Motors AB

Fabrikgatan 7, 54150

Skövde

Sweden

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1. Project Overview

Line 4 is a new Advance Motor's production line that will process the cylinder heads in the engine factory at Skövde. The line should be able to handle different models and variations of cylinder heads, for old and new engines. The line consists of 7 different operations that process the products sequentially. The supplier must deliver and assemble a cell carrying Operation 70 (OP70).

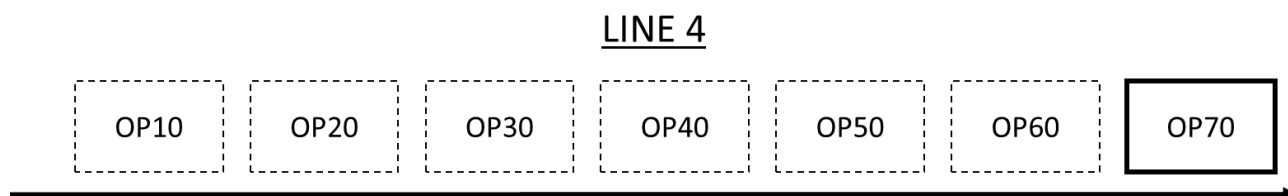


Figure 1: Line 4 overview

In OP70, the final operation in the line, products that have been rejected in the previous inspection operation are must be taken out from the line and palletized.

1.1. Time Plan

The following milestones have been set up for the delivery of OP70:

- M01 - Quote approved
- M02 - Design review
- M03 - FAT test
- M04 - SAT test

1.2. Delivery location

The cell is to be delivered to the processing plant, building B1 at the address below:

Fabrikgatan 7, 54150

Skövde

Sweden

2. Technical specifications

2.1. Description

All products arriving at OP70 will be marked as accepted or rejected in the previous inspection operation (OP60). Accepted products will be moved straight forward out from the cell, to the next line, while rejected ones should be extracted. The products will arrive from a conveyor line from OP060, and once in the extraction station, the rejected ones should be extracted, completely automatically and without human interaction, and placed in pallets (euro pallets). Different products should not be mixed in a pallet and the amount of products in a pallet cannot exceed 3 products.

A parallel conveyor system will handle the pallets, including necessary buffers and two stations, one on each end. This stations will be used for the removal of complete pallets and supply of empty ones. The extraction and supply of pallets will be carried out by an operator driving a forklift. The buffers will be included to allow flexibility in the pallet supply and extraction processes. While extracting and supplying pallets, the conveyors on which the pallets are placed should be safely stopped.

Just one operator should be able to handle the entire operation from the panel, so it should be placed close to the pallet extraction or supply stations. The operator will be advised using HMI devices when there is a need to supply empty pallets or remove complete ones. The operator will always confirm supply or removal of pallets, using HMI devices placed so they can be reached or actioned preferably from the forklift.

Operators safety should be ensure at all the time by using safety devices like light curtains and fences. Reset and start of the system by the operator should be as smooth as possible.

2.2. Performance and operating characteristics

The plant shall meet the following performance and operating characteristics:

- Real capacity must be at least 50 parts per hour.
- Availability must be at least 95%

2.3. Working flow

The picture bellow shows the suggested working flow of the extraction station of rejected products. Boxes show different states, while arrows show actions. The working cycle with a maximum period of 72 seconds restarts after the last actions.

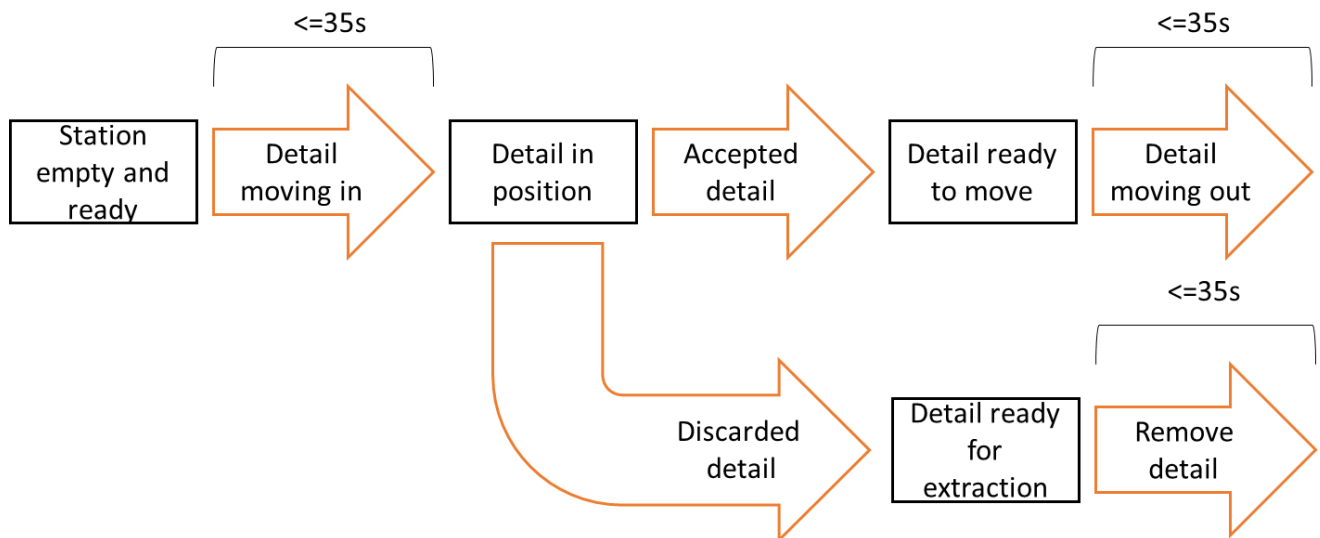


Figure 2: Operation working flow

To achieve the desired capacity, the time taken for moving products in and out to/from the station as well as removing them, should be no longer than 35 seconds respectively.

The parallel pallet handling system should be designed to ensure that empty pallets are always available for the extraction of rejected products.

2.4. CE-marking

The supplied cell shall be CE marked and all components incorporated must have CE marking.

2.5. Installation of electrical cabinet

For location of the cabinets see layout. Placement in separate rooms in a dirty environment if possible.

2.6. Equipment and maintenance

All movements in machines must have sensors in their end positions, for example cylinders. Servo shafts, etc.

All equipment in the installation should be accessible for maintenance. Required safety gates should be installed for safety areas divided by fences.

2.7. Products

Advance Motors AB manufactures engines of different sizes and variants of each of them. The plant should be able to handle the existing products and models described below. The cell should be designed and developed to handle new models in the future.

2.7.1. Product Models

Current models are:

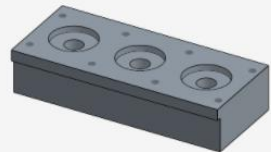
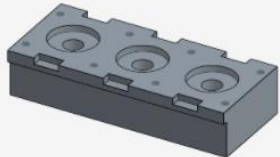
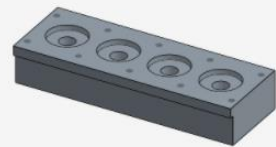
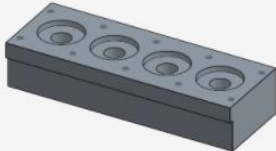
Name	Model	Size (l/w/h)	Product Number	Picture
Engine 1.6 Liter	MB16	700x300x150	1001605	
Engine 1.6 Liter Turbo	MB16T	700x300x150	1001610	
Engine 2.0 Liter	MB20	850x300x150	1002004	
Engine 2.4 Liter	MB24	850x300x170	1002407	

Figure 3: Motor models

The weight of the different models varies from 20 kg to 45 kg. CAD drawings for different models are included as attachment. Detailed 3D models can be delivered when the offer is approved and the privacy agreement is signed.

2.7.2. Batch size

Different models are produced in batches which sizes can vary from 20 to 100 products/batch.

2.7.3. Production data

Production data will be following each detail through the line and will be transferred between different PLCs. This production data includes 'Product Number', 'ID Number' and 'Status' information.

- Product Number (16 bits): MB16='1605', MB16T='1610', MB20='2004' and MB24='2407'.
- ID Number (32 bits): It is a unique number to identify each product (9 digits) = YYMMDDXXX (where XXX is the running number).
- Status (16 bits): Each bit in the status word is used to mark products status after different processes. OP060 will use bit 6 to mark the product as passed='1' or not passed='0' after inspection. Rest of the bits in the status word should not be changed, keeping their previous value before entering the operation.

2.8. Layout:

The delivered cell should be limited to the following dimensions:

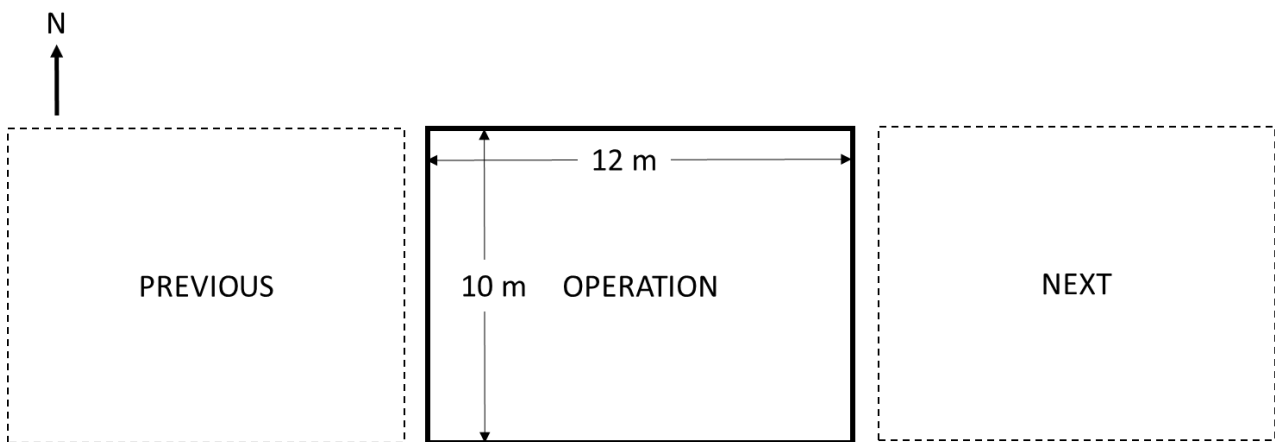


Figure 4: Dimensions of the cell

Detailed layout of the operation is shown below. Advance Motors will put Electric and Pneumatic air outlets at the selected point.

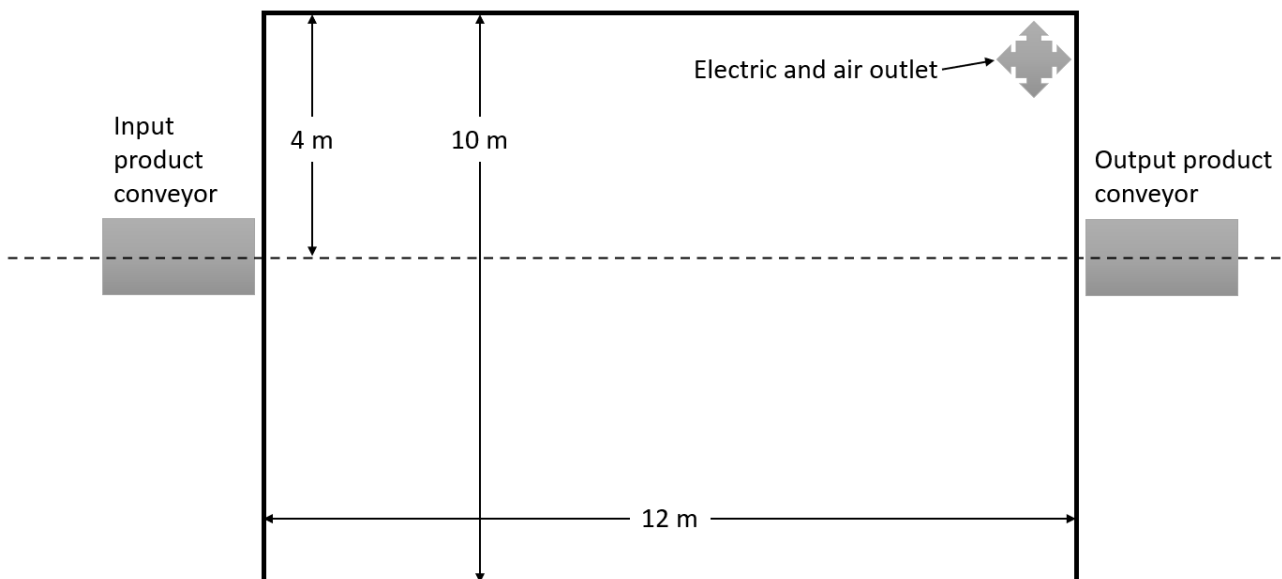


Figure 5: Detailed layout of the cell

3. Safety

The facility shall include all necessary safety equipment and safety features.

3.1. Emergency Stop (Optional)

Referred to an uncontrolled stop used in danger situations. Emergency stops in the form of pushbuttons should be on the main panel and at each door to the safety zone. Possibly more emergency stop pushbuttons may be required. When an emergency stop pushbutton is affected, all movements are stopped.

Which emergency stop zone is affected should be displayed on the main panel display with alarm text and graphic indication. Emergency stop reset is only done from the main panel.

3.2. Safety Stop (Optional)

The safety stop shall mean that personal safety is achieved upon entry into the machine. As with emergency stop, a fault-safe break (kat3 P1 d) applies to electromechanical movements, venting, but takes place under controlled conditions. Movements outside the protective gates should not normally be broken. Safety stop shall not rise any alarm for any part of the machine, only indicate non-acknowledged gate.

Safety gates may not be dependent on the lock magnet to be held in a closed position, a mechanical lock must keep the door closed.

Which safety gate that enabled the safety stop should appear on the main panel display with message text and graphic indication.

3.3. Access to safety zone (Optional)

Access is preceded by a manually requested access request. Cycle must stop as soon as possible, some exceptions may occur, for example threading, screwing and marking. Charging equipment/robot must run the completed part-sequence.

Stop position must be reached before the gates are unlocked. A light indicates when access is allowed.

In manual mode the entry request must always unlock the gate.

It must be possible to reach the entire machine/surface through safety gates.

3.4. Exit safety zone (Optional)

Leaving the safety zone and restart should be made according to the following routine:

- Reset Protection (Close gate)
- Acknowledged gate. (Blue button)
- Cycle start. (Green Button).

All gates must be acknowledged in a safety zone in order to start.

When restoring a gate, no movements may occur until the zone is acknowledged.

4. Control system and HMI

4.1. PLC

Delivered control PLC must be compatible with Codesys v3.5.

4.2. Robot

The robot (s) chosen by the supplier shall be from ABB.

4.3. Main operator panel

With the operator panel, the operator must be able to start and stop production in a smooth manner.



Figure 6: Operator main panel

Name/Type	Button Function	Light Function
EMERGENCY STOP (Normally closed push button)	The machine is stopped as quickly as possible and all electrical control signals are cut by hardware. (To be used only in emergencies.)	
CONTROL VOLTAGE ON (push button)	Connects control circuits in the machine.	Grey light solid when system power is on.
CONTROL VOLTAGE OFF (Push button)	Disconnects control circuits in the machine.	
MODE SELECTION MAN - AUTO (Switch)	Switch for operating mode selection. In "MAN" mode, the machine is set to manual mode. In "AUTO" mode, the machine is set to automatic mode.	
CYCLE START (Push button)	Press 3 seconds to start automatic operating mode.	Green cycle light flashing (1 Hz) when start of automatic cycle is possible. Solid when the automatic cycle is in progress.
CYCLE STOP (Push button)	Stops the automatic cycle in progress in an easy to start state (when equipment are idle).	
HALT (Normally closed push button)	All movements are stopped immediately. Auto sequence paused.	
RESET ALARM (Push button)	Acknowledges and erases alarms.	Red alarm light is flashing (1 Hz) when there are unacknowledged alarms Light is turned off when alarm is acknowledged.

Table 1: Main panel button and indicator functionality description

4.4. Gate panels

At each gate there must be a gate panel. Exception when main panel is close to it.

The following functions are controlled from the gate panel:



Figure 7: Safety gate panel

Name/Type	Button Function	Light Function
EMERGENCY STOP (Normally closed push button)	The machine is stopped as quickly as possible and all electrical control signals are cut by hardware. (To be used only in emergencies.)	
CYCLE START (Push button)	Press 3 seconds to start automatic operating mode.	Green cycle light flashing (1 Hz) when start of automatic cycle is possible. Solid when the automatic cycle is in progress.
ENTRY REQUEST (Normally closed push button)	The machine's movements are stopped after completion of the program step (Like Cycle Stop). And gate is unlocked.	Green cycle light flashing (1 Hz) when entry request is manifested and start of automatic cycle is possible. Solid when the automatic cycle is in progress.
RESET ALARM (Push button)	Acknowledges and erases alarms.	Red alarm light is flashing (1 Hz) when there are unacknowledged alarms

		Light is turned off when alarm is acknowledged.
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Table 2: Safety gate panel button and indicator functionality description

4.5. Light beacons

A light beacon should be used to visually provide the status of the cell, it should be positioned so that it is mostly visible inside and outside the cell. The following status colors should be placed and used in order:



Name/Type	Light Function
Red beacon	Flashing (1 Hz) when the alarm is unacknowledged. Solid when there is a remaining acknowledged alarm.
Yellow beacon	Flashing (1 Hz) when a warning is active.
Green beacon	Flashing (1 Hz) when start of automatic cycle is possible. Solid when the automatic cycle is in progress.
Blue beacon	Solid when the cell is waiting for a product.

4.6. HMI Displays

4.6.1. General

Current alarms and warnings are displayed in all images.

If a function key has the same function in more than one image, the function key must keep its place in all images where the function is located.

It should be possible to change the mode of operation regardless of the image displayed on the screen.

The display must be switched off or tinted down after 30 minutes of inactivity.

It should be possible to display all images regardless of the current mode of operation.

When attempting to enter incorrect values, selection of unauthorized functions, start of motionless movements, etc., an explanatory error message should be displayed.

Some features should be password protected. Password usage is done in consultation with Advance Motors.

Pictures that are required should be provided with some help such as Explanatory text or shortcut to help function.

There should be an option to alternatively switch between symbols and absolute addressing in all applicable images.

Below are the standard images that will be found in the HMI. Additional pictures to realize described features may be required depending on equipment. Note that only image features are described in this chapter. HMI structure should follow chapter HMI structure and sample images below.

4.6.2. Menu

The menu is displayed in the bottom of the HMI display. The selected page is highlighted.

All main pages in the HMI panel should be directly available from the main menu.

OVERVIEW	PROCESS	MANUAL	PRODUCTION	DIAGNOSTICS	HMI
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Table 3: Operator display page menu

4.6.3. General Overview

The most used information will be collected in a single overview image. The picture should also show the current detail in the equipment.

The image must contain a graphical machine layout to show current state of the equipment and indication of alarm part in the station.

The image must contain all emergency stops and protection gates in their location and their current status.

4.6.4. Process layout

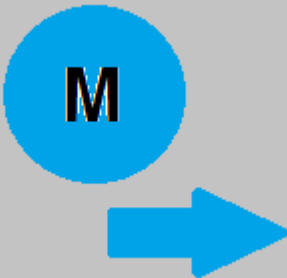
A graphical image of the equipment (3d drawing preferred) that shows where sensors and actuators are located and their current status: which engine/engines are working, which valves are open/closed, even levels and temperatures should be displayed. The components shall be described with reference designation and text comment.

Sensor status indicated by:

- Green = Affected in normal mode
- Yellow = Affected in not normal mode
- Red = Error detected
- Blank = Not affected

Actuator status indicated by:

- Blue = Active
- Red = Error detected
- Blank = Not active

Motor example:		
Not Active	Active	Error
		
Cylinder example:		
Double acting cylinder with two positions: Retracted (Normal position), yellow sensor, and extracted, green sensor.		
Not Active	Active	Error
		

4.6.5. Manual operation

Movements and functions shall be performed with function keys located on the left and right sides of the image. The movements and functions shall be so structured that their starting position is grouped together. Information must be provided regarding position achieved, movement is in progress and if movements are possible to perform, see explanation below. On the manual page for axle driving, the shaft position should also be displayed.

Explanation for manual movements:

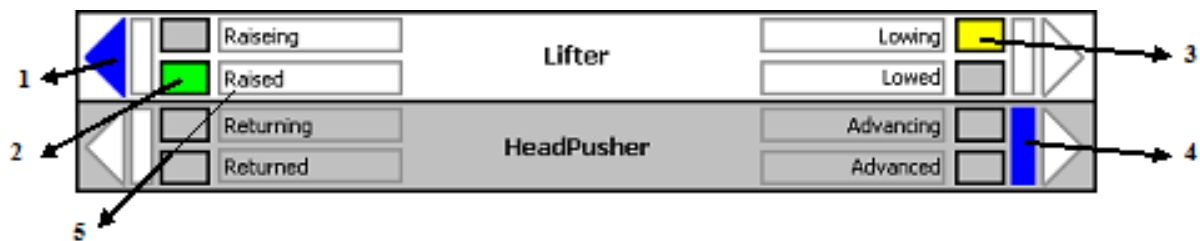


Figure 8: Manual movement example

No.	Information
1	Blue triangle, possible to drive motion. Gray triangle, motion blocked.
2	Position reached
3	PLC output activated
4	Function key for movement is pressed.
5	The normal position of the functions should be grouped on the same page.

Table 4: Manual movement functionality description

4.6.6. Production

Handling of production-related data.

4.6.6.1. Part counters

There must be 2 counters that calculates total number of executed parts, one that cannot be reset and another that can be reset.

4.6.6.2. Cycle time

The cycle time of the cell must be measured from part out to next part out.

4.6.7. Diagnostics

4.6.7.1. Current alarms

The alarms will be presented in time with the oldest alarm at the top. Alarms should only be active as long as the cause of the alarm persists. Acknowledge required.

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4.6.7.2. Alarm history

The alarms must be stored in a log and presented in time with the youngest alarm at the top.

4.6.7.3. Current warnings / messages

Message/warning should be presented in time sequence with the oldest message at the top. Message and alerts should only be active as long as the reason for the message/warning persists. Acknowledge is not required.

4.6.7.4. Warnings history

Warnings must be saved in a log and presented in time with the youngest message at the top. Message history may not be included.

4.6.8. HMI (and gate panels)

The main operator HMI is presented together with gate panels.

5. Functional description

5.1. Machine Startup

Machine boot should be as follows:

- Control voltage ON
- Reset of security/alarms and emergency shutdown may be required.

5.2. Operating modes from the main panel

Change between operating modes should have immediate effect. For example, if switching from automatic to manual mode is done during automatic cycle, this will result in an immediate shift to manual mode and the cycle will be interrupted.

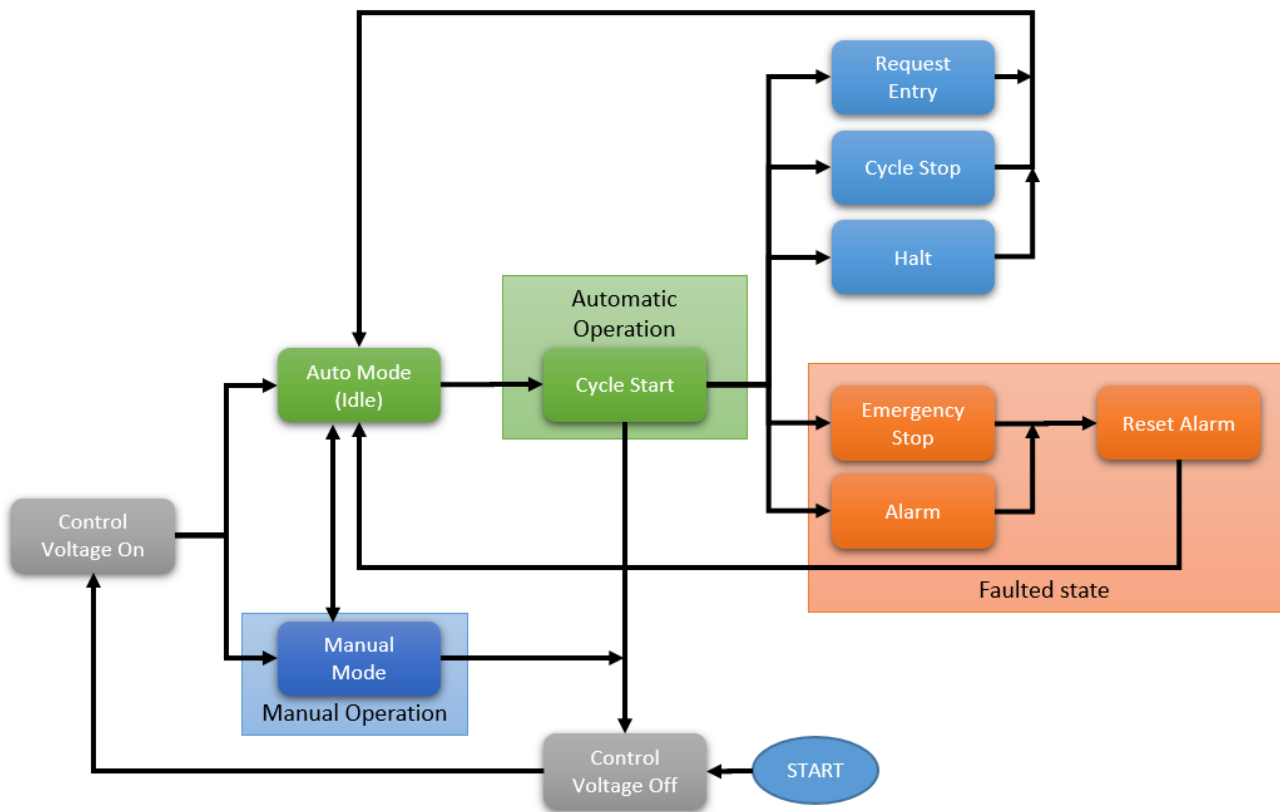


Figure 1 Operational modes and states architecture

5.2.1. Manual Operation

The movements are selected and ordered from the panel. The movement should be canceled as soon as the button is released.

Manually maneuver of movements should be allowed only in this mode.

For functions that do not work against fixed stop modes, for example conveyor motors, movement is stopped when the position sensor is activated or the button released. After that the actuators must be re-actuated in order to continue the movement.

5.2.2. Manual Assignment

There should be the possibility to drive different manual assignments, for example, service, service position for grippers, adjustments and settings etc. for robots. In addition, all robot missions can be run separately.

5.2.3. Automatic Operation

The machine repeats complete work cycles continuously. By pressing cycle stop the machine can be stopped in an easy to restart state. Immediate stopping can be performed by pressing the halt button and in emergencies pressing the emergency stop.

5.2.4. Faulted State

When an emergency stop is pressed or an alarm is activated the system stops immediately. Information about the stop is displayed in the HMI diagnostics page. Alarms must be acknowledge in the HMI and in order to restart the automatic mode the alarm reset button has to be pressed.

5.3. Restart

It should be possible to take a cycle back to the last step before machine was stopped.

6. Alarm functions: Faults, warnings and messages

The basic principle is that all error states should generate an alarm that is presented in plain text on the operator panel.

A complete alarm text presented to the operator must contain: Alarm number, identifier and a description of the event. It will include I/O address if it is related to an I/O device.

Components and devices, as well as software functions such as transporters, sequences, movements, engine protection, fuses, signal exchange with other equipment, etc. shall be monitored individually. In abnormal situations, alarms must be generated and presented to the operator.

Follow-up alarms caused by a previous malfunction should, if possible, be prevented. Timeout alarm should not be given if time expires due to consequent errors.

Faults must be active until the cause of the problem has been resolved and the operator has acknowledge it. The fault acknowledge and the reset of the text in the display must be done with the same button.

When a fault occurs, an alarms should be generated on the panel and make the red-light button flash. When the operator acknowledges that is aware of the alarm, the lamp will remain solid if the fault cause remains.

A fault in a station may only stop the wrong part of the station. The rest of the station and the machine should continue to work if possible.

In manual mode or with an open gate, timeout, sequence, sensor pair, frequency converter etc. faults must be avoided. To minimize false alarms during service.

6.1. Alarm levels

Alarms should be divided into different levels depending on the severity of the error:

- A. Fault: Immediate stop
- B. Fault: Stop after completion of the partial cycle (work on as long as possible).
- C. Warning - Do not stop the machine.
- D. Message to operator - Do not stop the machine.

6.1.1. Faults (level A / B)

Faults must be active until the cause has been resolved and the operator has acknowledge it.

Black text with red background on HMI.

6.1.2. Warnings (Level C)

Warnings reset themselves when the cause for the warning has ceased. Use of warnings should be carefully thought out to avoid damage to the machine, etc.

Black text with yellow background on HMI.

6.1.3. Messages (Level D)

Messages should be created when no prerequisite for operating mode or movement is missing while the operator attempts to activate the operation. When attempting to enter incorrect values, selecting unauthorized functions, etc., an explanatory error message will also be displayed. Messages should also be created by conscious action such as gate opening or shutdown of work switches etc.

Black text with blue background on HMI.

7. Deliverables

The supplier should provide the following deliverables.

- **System design:** A detailed functional description document with all necessary diagrams and flow charts.
- **Technical documentation:** Including:
 - Network diagram
 - A complete set of Electric and Pneumatic drawings.
 - A list of all electric and pneumatic components, including control systems, safety elements, sensors, contactors, drives, motors, valves, etc.
 - I / O List
- **User and maintenance Manual:** An operator manual for handling the cell in normal production, alarm handling and troubleshooting information.
- **FAT/SAT Protocol:** A list of test to be carried to verify and validate supplied cell, according to the specifications in this document and the system design document.
- **Control System and HMI programs:** PLC and HMI programs with documentation.

8. FAT/SAT

The supplier, together with Advance Motors, will review documentation and installation, including:

- Equipment components and documentation
- Safety components
- User and maintenance manual
- Technical drawings
- Controller and HMI display code

Finally, previously agreed FAT/SAT protocol will be validated.

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9. Delivery

Final visual inspection of the cell and marking will be done. The supplier shall provide the last version of the technical documentation. After all integration, safety and functional tests, driving capability with operators will be tested on site, at Advance Motors.

Training will be supplied for:

- Operators.
- Electrical Maintenance, including program review of PLC and Robot.
- Mechanical Maintenance

10. Payment Terms

Billing takes place after approved SAT, milestone M4. Payment is made 90 days from invoice date.

11. Attachment A: Product CAD Drawings

